UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 10/01/2019*

EXERCISE

It is given a wide-sense stationary (w.s.s.) Auto-Regressive (AR) process $S[n]$ of order 2, $S[n] = -0.7 \cdot S[n-1] + 0.18 \cdot S[n-2] + W[n]$, where $W[n]$ is the zero-mean white innovation process with power σ_W^2.

(1) Derive the system function $L(z)$ of the innovation filter, its zeros and poles, and determine if $S[n]$ is a low-pass, high-pass, or band-pass process. Then, derive the power spectral density (PSD) $S_S\left(e^{j2\pi f}\right)$ and the autocorrelation function (ACF) $R_S[m] = E\{S[n]S[n+m]\}$ of the process $S[n]$ and plot them. Finally, derive the power of the innovation process σ_W^2 in such a way that $\sigma_S^2 = 1$.

(2) Derive the 1-step optimal linear predictor of order *P*=1 of the process $S[n]$, i.e. the linear minimum mean square error (LMMSE) estimator of $S[n]$ given the observed data $S[n-1]$. Derive its mean square error (MSE) $\sigma_\varepsilon^2 \triangleq E\{\varepsilon^2[n]\} = E\left\{\left(S[n] - \hat{S}[n]\right)^2\right\}$.

(3) Derive the 1-step optimal linear predictor of order *P*=2 of the process $S[n]$, i.e. the LMMSE estimator of $S[n]$ given the observed data $S[n-1]$ and $S[n-2]$ and its MSE. Explain what happens if we further increase *P*.

Hint: $\rho_S[1] = -0.854, \quad \rho_S[2] = 0.778$, where $\rho_S[m]$ is the normalized ACF.

Solution.

(1) $S[n]=-0.7\cdot S[n-1]+0.18\cdot S[n-2]+W[n]$

$$S(z)=-0.7z^{-1}S(z)+0.18z^{-2}S(z)+W(z) \quad \Rightarrow \quad L(z)=\frac{S(z)}{W(z)}=\frac{1}{1+0.7z^{-1}-0.18z^{-2}}$$

$$L(z)=\frac{1}{\left(1-0.2z^{-1}\right)\left(1+0.9z^{-1}\right)}=\frac{z^2}{\left(z-0.2\right)\left(z+0.9\right)}$$

The innovation filter has 2 zeros at $z=0$, 1 pole at $z=p_1=0.2$ and 1 pole at $z=p_2=-0.9$. The innovation filter is a high-pass filter since the dominant pole $p_2=-0.9$ is real and negative, and so the process is a high-pass process.

The PSD and the ACF are given by:

$$\begin{aligned}S_S\left(e^{j2\pi f}\right)&=\sigma_W^2\left|L\left(e^{j2\pi f}\right)\right|^2=\frac{\sigma_W^2}{\left|1-0.2e^{-j2\pi f}\right|^2\left|1+0.9e^{-j2\pi f}\right|^2}=\frac{\sigma_W^2}{\left[1.04-0.4\cos\left(2\pi f\right)\right]\left[1.81+1.8\cos\left(2\pi f\right)\right]}\\&=\frac{\sigma_W^2}{1.04\cdot 1.81+\left(1.04\cdot 1.8-1.81\cdot 0.4\right)\cos\left(2\pi f\right)-0.4\cdot 1.8\cos^2\left(2\pi f\right)}\\&=\frac{\sigma_W^2}{1.8824+1.1480\cdot\cos\left(2\pi f\right)-0.7200\cdot\cos^2\left(2\pi f\right)}\\&=\frac{\sigma_W^2}{1.5224+1.1480\cdot\cos\left(2\pi f\right)-0.3600\cdot\cos\left(4\pi f\right)}\end{aligned}$$

$R_S[m]=E\left\{S[n]S[n+m]\right\}=R_S[0]\rho_S[m]$, where the normalized ACF $\rho_S[m]$ is the solution of the linear difference equation: $\rho_S[m]+0.7\cdot\rho_S[m-1]-0.18\cdot\rho_S[m-2]=0$, that is:

$\rho_S[m]=A\cdot\left(0.2\right)^{|m|}+B\cdot\left(-0.9\right)^{|m|}$,

where the two constants A and B are derived from the initial conditions:

C.1) $\rho_S[0]=A+B=1$

Calculating the normalized ACF for m=1 we get: $\rho_S[1]+0.7\cdot\rho_S[0]-0.18\cdot\rho_S[-1]=0$, and keeping in mind that $\rho_S[0]=1$ and $\rho_S[1]=\rho_S[-1]$, we finally get: $\rho_S[1]=\frac{-0.7}{1-0.18}=-0.854$

Hence, the 2nd initial condition is:

C.2) $\rho_S[1]=0.2A-0.9B=-0.854$

Solving the linear system:

$$\begin{cases} A+B=1 \\ 0.2A-0.9B=-0.854 \end{cases}$$

we finally get: A=0.042, B=0.958.

$$\rho_S[m]=0.042\cdot(0.2)^{|m|}+0.958\cdot(-0.9)^{|m|}$$

The power of the process $S[n]$ can be found as follows:

$$\begin{aligned} R_S[0]&=E\left\{S^2[0]\right\}=E\left\{\left(-0.7\cdot S[n-1]+0.18\cdot S[n-2]+W[n]\right)^2\right\} \\ &=(-0.7)^2R_S[0]+(0.18)^2R_S[0]+2(-0.7)(0.18)R_S[1]+\sigma_W^2 \end{aligned}$$

$$R_S[0]-(-0.7)^2R_S[0]-(0.18)^2R_S[0]-2(-0.7)(0.18)R_S[0]\rho_S[1]=\sigma_W^2$$

$$R_S[0]=\frac{\sigma_W^2}{1-(-0.7)^2-(0.18)^2-2(-0.7)(0.18)(-0.854)}=\frac{\sigma_W^2}{0.262}$$

$\sigma_S^2=R_S[0]=\frac{\sigma_W^2}{0.262}$, quindi $\sigma_S^2=1 \;\; \rightarrow \;\; \sigma_W^2=0.262$

In summary the ACF and the PSD are given by:

$$R_S[m] = R_S[0]\rho_S[m] = 0.042 \cdot (0.2)^{|m|} + 0.958 \cdot (-0.9)^{|m|}$$

$$S_S\left(e^{j2\pi f}\right) = \frac{0.262}{1.5224 + 1.1480 \cdot \cos(2\pi f) - 0.3600 \cdot \cos(4\pi f)}$$

Plots of the ACF and of the PSD are reported at the end of this document.

(2) $\hat{S}[n] = h[1]S[n-1] + b$

Since $S[n]$ is a zero mean process we have that: $E\{\varepsilon[n]\} = 0 \quad \Rightarrow \quad b = 0$.

$$\hat{S}[n] = h[1]S[n-1]$$

$$E\{\varepsilon[n]S[n-m]\} = 0, \quad m = 1$$

$$E\left\{\left(S[n] - \hat{S}[n]\right)S[n-1]\right\} = 0$$

$$E\{(S[n] - h[1]S[n-1])S[n-1]\} = 0$$

$$R_S[1] - h[1]R_S[0] = 0$$

$$h[1] = \frac{R_S[1]}{R_S[0]} = \rho_S[1] = -0.854$$

$$\hat{S}[n] = -0.854S[n-1]$$

$$\sigma_\varepsilon^2 = E\left\{\varepsilon^2[n]\right\} = E\left\{S[n]\left(S[n] - \hat{S}[n]\right)\right\} = E\left\{S[n](S[n] + 0.854S[n-1])\right\}$$

$$= R_S[0] + 0.854R_S[1] = 1 + 0.854(-0.854) = 0.271$$

(3) $\hat{S}[n] = h[1]S[n-1] + h[2]S[n-2] + b$

Again, since $S[n]$ is a zero mean process we have that: $E\{\varepsilon[n]\} = 0 \quad \Rightarrow \quad b = 0$.

$$E\left\{\varepsilon[n]S[n-m]\right\}=0, \quad m=1,2$$

$$E\left\{\left(S[n]-\hat{S}[n]\right)S[n-m]\right\}=0, \quad m=1,2$$

$$E\left\{\left(S[n]-h[1]S[n-1]-h[2]S[n-2]\right)S[n-m]\right\}=0, \quad m=1,2$$

$$R_S[m]-h[1]R_S[m-1]-h[2]R_S[m-2]=0, \quad m=1,2$$

$$\begin{cases} h[1]R_S[0]+h[2]R_S[-1]=R_S[1] \\ h[1]R_S[1]+h[2]R_S[0]=R_S[2] \end{cases}$$

$$\begin{bmatrix} R_S[0] & R_S[-1] \\ R_S[1] & R_S[0] \end{bmatrix}\begin{bmatrix} h[1] \\ h[2] \end{bmatrix}=\begin{bmatrix} R_S[1] \\ R_S[2] \end{bmatrix}$$

$$R_S[0]=1, \quad R_S[1]=R_S[-1]=-0.854 \quad R_S[2]=0.042\cdot(0.2)^{|2|}+0.958\cdot(-0.9)^{|2|}=0.778$$

$$\begin{bmatrix} 1 & -0.854 \\ -0.854 & 1 \end{bmatrix}\begin{bmatrix} h[1] \\ h[2] \end{bmatrix}=\begin{bmatrix} -0.854 \\ 0.778 \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix}=\begin{bmatrix} -0.700 \\ 0.180 \end{bmatrix}$$

$$\hat{S}[n]=-0.7\cdot S[n-1]+0.18\cdot S[n-2]$$

$$\sigma_\varepsilon^2=E\left\{\varepsilon^2[n]\right\}=E\left\{\left(S[n]-\hat{S}[n]\right)^2\right\}=E\left\{W^2[n]\right\}=\sigma_W^2=0.262<0.271$$

Increasing *P,* the LMMSE estimator does not change ($h[n]$=0 for n>2) and so the MSE does not decrease.

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Figures: